

Complex Analysis – 7

Конформные отображения и дробно-линейные преобразования

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Möbius transformations, the cross-ratio and the circle-preserving property; automorphisms of the sphere, the plane, the disc and the half-plane; the standard dictionary z^α , e^z , $\log z$, $\frac{1}{2}(z + z^{-1})$; the Riemann mapping theorem and its uniqueness normalisation; the Schwarz reflection principle; the conformal invariant of an annulus (модуль кольца).

1. Prove that every automorphism of the upper half-plane is $z \mapsto \frac{az+b}{cz+d}$ with real a, b, c, d and $ad - bc > 0$, and classify Möbius transformations into elliptic, parabolic and hyperbolic by the trace of the corresponding matrix.
2. (S.T. Yau CSMC 2018) Let $D \subsetneq \mathbb{C}$ be simply connected and $z_0 \in D$. Prove that there exists a univalent $f: D \rightarrow \mathbb{D}$ with $f'(z_0) > 0$ *without* appealing to the Riemann mapping theorem.
3. (S.T. Yau CSMC 2022) Let $\Omega \subsetneq \mathbb{C}$ be simply connected, $z_0 \in \Omega$, and let $F = \{f: \Omega \rightarrow \mathbb{D} \text{ holomorphic, injective, } f(z_0) = 0\}$. Explain in detail the proof of the Riemann mapping theorem based on maximising $|f'(z_0)|$ over F .
4. Prove the Schwarz reflection principle, and use it to show that a conformal map of the upper half-disc onto the upper half-plane extends to a conformal map of the disc onto the plane minus two slits.
5. Map conformally onto the unit disc: (a) the strip $\{0 < \operatorname{Im} z < \pi\}$; (b) the half-strip $\{0 < \operatorname{Im} z < \pi, \operatorname{Re} z > 0\}$; (c) the complement of the segment $[-1, 1]$.

6. Study the Joukowski map $w = \frac{1}{2}(z + \frac{1}{z})$: find the images of circles $|z| = r$ and of rays, determine where it is conformal and univalent, and explain how it produces the exterior of an ellipse.
7. (S.T. Yau CSMC 2018, team) Let Ω be a simply connected domain, $z_0 \in \Omega$, and let $G(z, z_0)$ solve the Dirichlet problem with boundary values $\log |\zeta - z_0|$; put $g(z, z_0) = G(z, z_0) - \log |z - z_0|$. Let $w = f(z)$ be the conformal map of Ω onto $\mathbb{D}$ with $f(z_0) = 0$. Show that $|f(z)| = e^{g(z, z_0)}$ up to the standard normalisation, i.e. that the Green's function determines the Riemann map.
8. Prove that the modulus $\frac{1}{2\pi} \log \frac{r_2}{r_1}$ of an annulus is a conformal invariant and that two annuli are conformally equivalent exactly when their moduli agree; deduce that the punctured disc and the punctured plane are not conformally equivalent.
9. Prove that a conformal map of $\mathbb{D}$ onto $\mathbb{D}$ fixing the origin with positive derivative is the identity, and use it to establish the uniqueness part of the Riemann mapping theorem.
10. State the uniformisation theorem for the thrice punctured sphere: the universal cover of $\mathbb{C} \setminus \{0, 1\}$ is the disc. Deduce Picard's little theorem from it, and explain why an entire function omitting two values would lift to a bounded entire function.